ReturnLens

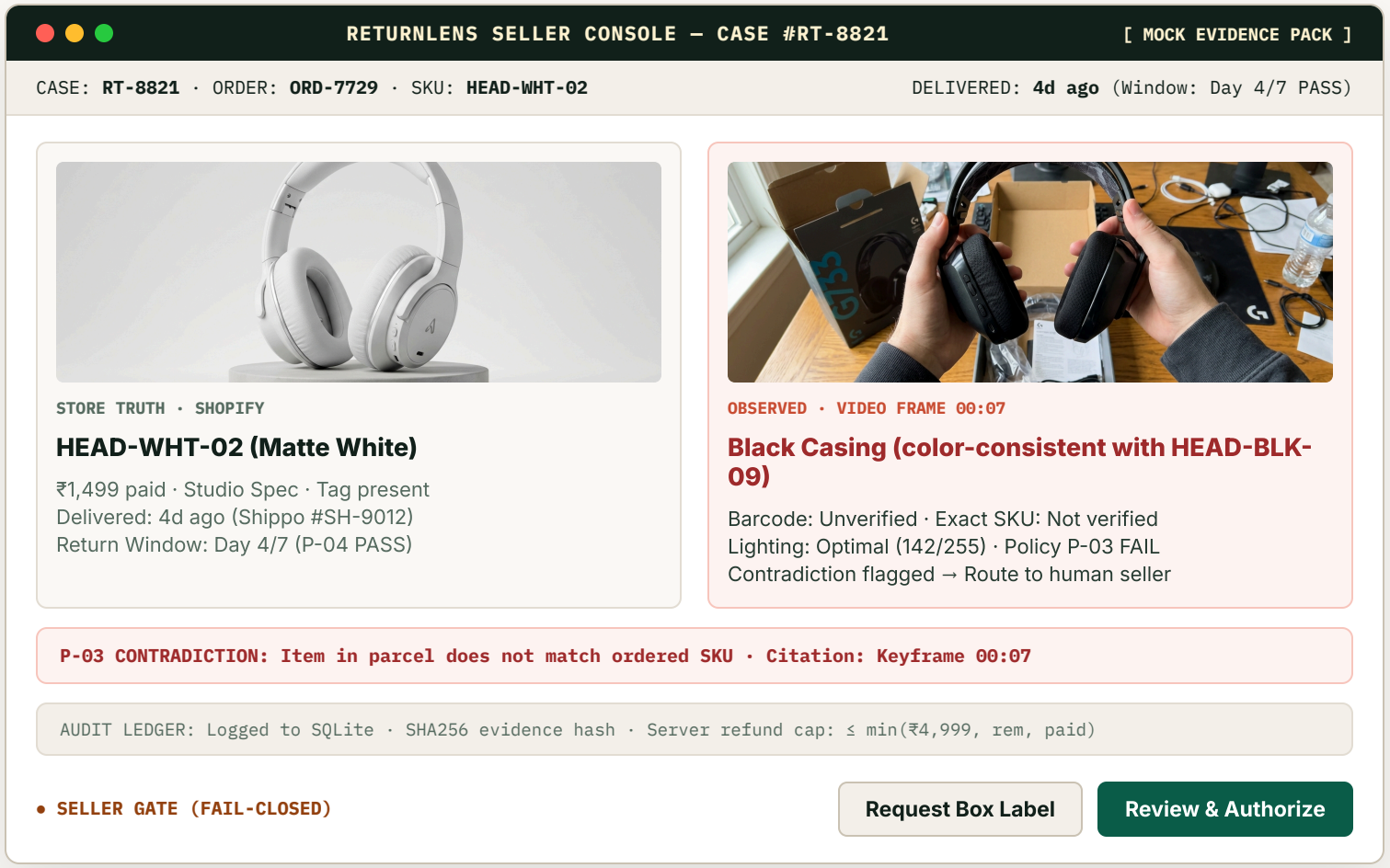
Every return decision, backed by evidence.

ReturnLens unifies unboxing video evidence, store truth, and policy checks into a 1-click decision console for solo merchants.

CORE ARCHITECTURAL PRINCIPLE

“Models observe. Code verifies. Seller authorizes.”

Scope: Return triage + order status · Mock fixtures · Never auto-refunds
Guardrail: Server cap ≤ min(₹4,999, remaining, paid) · Hash-bound



THE PROBLEM · SOLO MERCHANT BOTTLENECK · 4 SOURCES

One return. Four disconnected systems. Zero shared record.

Solo merchants cross-reference tabs, video footage, and policy rules manually before moving money.

SOURCE 01 **BUYER MESSAGE**
Subjective text & unindexed video claim in chat

SOURCE 02 **ORDER RECORD**
Shopify manifest: HEAD-WHT-02, Matte White, ₹1,499

SOURCE 03 **COURIER STATUS**
Shippo scan: delivered 4 days ago (Day 4 of 7)

SOURCE 04 **RETURN POLICY**
7-day defect window; security packaging tag rule

THE CRITICAL INFORMATION GAP

“What verifiable evidence justifies moving this money?”

Four disconnected systems. One solo seller making high-stakes financial decisions across browser tabs.

Without an auditable evidence pack, the merchant either refunds blind (losing COGS and 2-way freight) or disputes unfairly (destroying customer trust and merchant ratings).

INDUSTRY FRAUD BASELINE (DEFINITIONS DIFFER · NOT DIRECTLY COMPARABLE)
- 9.0% Return Fraud (NRF 2025): Strict criminal fraud only (unboxing swaps, empty boxes).
- 15.14% Abuse (Appriss/Deloitte 2024): Criminal fraud combined with policy abuse (wardrobing).

SOLO MERCHANT ASYMMETRIC RISK
- Wrong refund = Lost product COGS + 2-way shipping costs (unrecoverable margin loss).
- Unfair denial = Ruined customer trust, chargeback penalties & degraded seller account health.

STAR DEMO · CASE B · CORE BENCHMARK

The order says white. The video shows black.

ReturnLens flags the contradiction — the seller decides.

ORDER MANIFEST · STORE CATALOG TRUTH (#ORD-7729)



SKU: HEAD-WHT-02 · Spec: Matte White Casing

Price: ₹1,499 paid · Delivered: 4d ago (Shippo #SH-9012)
Return Window P-04 (7-Day Policy): PASS (Day 4 of 7)

≠

MOCK EVIDENCE PACK · VIDEO CITATION (FRAME 00:07)



Observed: Black Casing (color-consistent with HEAD-BLK-09)

Exact SKU: Not verified (Barcode unverified) · Policy P-03 FAIL
Lighting: Optimal (142/255) · REC 00:07 / 00:24 (Mock Keyframe)

POLICY P-03 RESULT: ESCALATE TO SELLER GATE (FAIL-CLOSED)

Guardrail: 0 < amt ≤ min(₹4,999, rem_ref, paid) · Never auto-refunds

1. Accept as warehouse mis-shipment
2. Request photo of outside-box shipping label
3. Hold return for physical parcel inspection

SYSTEM ARCHITECTURE · FAIL-CLOSED GOVERNANCE

Models observe. Code verifies. Seller authorizes.

Exactly 3 base model calls per triage case. Weak evidence escalates. Never releases automated refunds.

1. Planner	2. Tools & Vision	3. Policy Engine	4. Critic	5. Seller Gate	6. Audit Ledger
Model Call 1	Model Call 2	Deterministic Code	Model Call 3	Human Merchant	SQLite / JSON
Proposes typed JSON actions	FFmpeg keyframes + vision	Evaluates rules P-01–P-08	Verifies cited timestamps	Fail-closed 1-click auth	Append-only signed audit

CALL BUDGET & MODEL RESTRICTION

Exactly 3 Model Calls

1. Planner proposes typed tool actions.
2. Vision model inspects server-extracted keyframes.
3. Critic evaluates citations against policy.

MODELS NEVER ACCESS PAYMENT APIS DIRECTLY.

STATEFUL RESUMPTION (CASE C)

Clarification & Re-Film Loop

Low-light video (28/255 brightness) triggers ask_for_evidence().

State saves to WAITING_FOR_EVIDENCE → Resumes on upload.

TERMINATION RULE: After 1 retry, escalates to seller with both attempts logged (never auto-denies).

SECONDARY PS-04 ROUTE

Tiny PS-04 Order Route

For routine "Where is my order?" inquiries, the agent bypasses vision models entirely:

Store API → Courier API → Status Reply

Zero vision token waste · Instant response.

FAIL-CLOSED FINANCIAL GATE: requires_seller_approval: true · Approval binds case + evidence hash + policy version.

Server Cap: 0 < amount <= min(₹4,999, remaining_refundable, order_paid) · Revalidation check before draft release.

EVALUATION SUITE · 4 CONTROLLED SCENARIOS

Different evidence. Different action.

Every case produces an auditable mock evidence pack. No automatic policy approvals.

CORE BENCHMARK SCENARIO · P-03 CONTRADICTION

Case B: Color Contradiction

The order says White. Unboxing video frame 00:07 shows Black.

MOCK EVIDENCE PACK · FRAME 00:07

Manifest: HEAD-WHT-02 White (₹1,499)

Observed: Black casing (color-consistent)

Barcode: Unverified · Exact SKU: Not verified

P-03 FAIL → No label, no refund draft



POLICY ACTION: ESCALATE TO SELLER GATE (FAIL-CLOSED)

PRESERVED SELLER OPTIONS IN DECISION CONSOLE:

① Accept warehouse mis-shipment

② Request photo of outside-box shipping label

③ Hold return for physical parcel inspection

Financial Guardrail: 0 < amount ≤ min(₹4,999, rem, paid) · Binds evidence hash

Case A: Valid Return (ORD-1042) PREPARE SELLER REVIEW

Evidence: ORD-1042 · ₹1,499 · SKU matches catalog · Packaging tag present · Within 7-day return window.

Outcome: Mock shipping label created · Refund draft placed in pending_return_receipt (not executed).

Case C: Low-Light Video (Re-Film Loop) ASK & RESUME

Evidence: 4-second dark unboxing video · Brightness score 28/255 · Only 2 usable keyframes extracted.

Outcome: Agent issues re-film checklist · State saved to WAITING_FOR_EVIDENCE · Resumes on upload (after 1 retry, escalates with both attempts logged; never auto-denies).

Case D: Continuous Unboxing Sequence INSPECT SEQUENCE

Evidence: Single continuous take from opening sealed parcel to displaying label in one sequence.

Outcome: Inspects sequence continuity and flags abrupt transitions. Does not claim proof of identical item.

24-HOUR PROTOTYPE PLAN · TEAM ROOKIE (3 MEMBERS)

A credible prototype in 24 hours.

Strict scope boundaries and single-threaded ownership make the prototype buildable in one day.

00 – 06h · PHASE 01

Foundation

Mock Store/Shippo APIs · LangGraph state schema · Policy P-01 to P-08

06 – 12h · PHASE 02

Evidence Pipeline

FFmpeg ~1fps extractor · Vision prompt schemas · Seller Console UI mock

12 – 18h · PHASE 03

Integration

Case C pause/resume loop · 3 core cases (A–C) + Case D sequence test · Critic review

18 – 24h · PHASE 04

Validation & Safety

Fail-closed authorization tests · Boundary QA · Video demo recording

PERSON A · BACKEND, POLICY & AUDIT

State Machine & Policy Engine

Stack: LangGraph · Pydantic · SQLite · SHA256

- Policy P-01 to P-08: Deterministic code rules (window, SKU, tag, refund cap)
- Critic Agent: Adversarial check against cited keyframe timestamps
- Audit Ledger: Append-only JSON log & cryptographic approval binding
- Financial Cap: Server calculation min(₹4,999, remaining, paid)

PERSON B · VISION & CONSOLE

Video Pipeline & Seller UI

Stack: FFmpeg · Hosted vision model (unpinned) · HTML/CSS/SVG

- Keyframe Pipeline: Uniform ~1fps sampling & brightness scoring (Case C sample: 28/255)
- Structured Vision: Observation schema with timestamp citation metadata
- Seller Console: 1-click decision console with mock evidence pack
- Grounding Guard: Enforces unverified barcode vs color distinction

PERSON C · INTEGRATION & QA

Scenarios, Lifecycle & QA

Stack: Pytest · Mock Fixtures · Automated Smoke QA

- Benchmark Cases: Staged data for Cases A–C + Case D sequence test
- Case C Lifecycle: Re-film loop in WAITING_FOR_EVIDENCE (1 retry)
- Fail-Closed Testing: Verifies zero automated refunds & approval gates
- Automated QA: Extended smoke runner (10+ tests), end-to-end event bus & demo recording

MUST DELIVER (CORE PROTOTYPE)

Mock APIs · Policy engine · Cases A–C + Case D sequence test · Approval binding · Seller gate · Tiny order route

CUT FIRST IF BEHIND

Advanced video forensics (Case D stretch) · WhatsApp bot · Exchanges · Multi-seller auth · Audio analysis · Public cloud deployment

FINANCIAL CEILING FORMULA

```
0 < amt <= min(₹4,999, rem_ref, paid)
```